

Trainings ▾

General Information

Low Mount

A small amount of air could possibly be heard escaping from the turbocharger control valve during the turbocharger actuator test. This is a normal condition for the valve to achieve output regulation pressure.

Do **not** replace the turbocharger control valve for this condition.

The valve should **not** leak air with key on and engine off. If the valve leaks air with key on and engine off, the valve is damaged.

Remove

High Mount

Verify the keyswitch is in the OFF position.

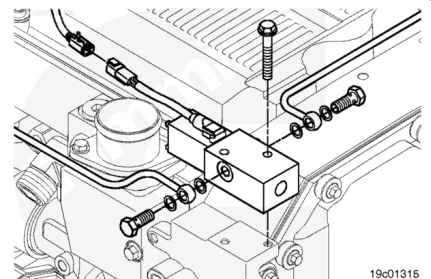
Bleed the air from the turbocharger control valve supply line.

Remove the air lines to the turbocharger actuator and the air supply.

Disconnect the electrical harness from the turbocharger control valve.

Remove the two mounting capscrews.

Discard the sealing washers.

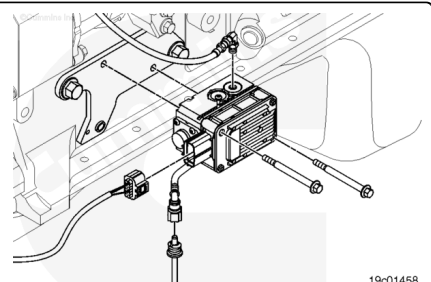


Low Mount

Verify the keyswitch is in the OFF position.

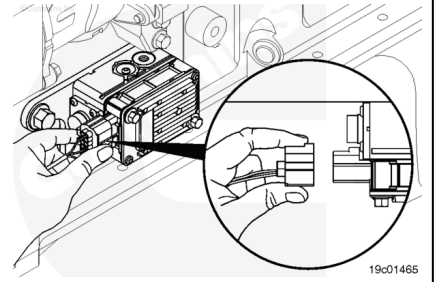
Bleed the air from the turbocharger control valve supply line.

Remove the air lines to the turbocharger actuator and the air supply.



⚠ CAUTION ⚠

Do not pull on the wiring harness. Damage to the harness will result.

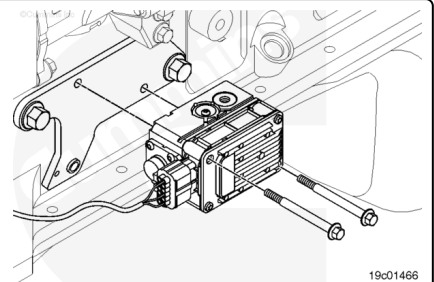


Disconnect the electrical harness from the turbocharger control valve.

To remove the connector, place the forefinger on the connector release with the thumb on the connector, while pressing the release mechanism with the forefinger as shown.

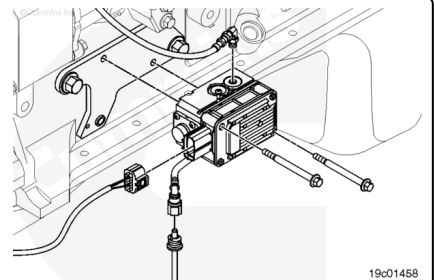
Push in on the release mechanism and pull firmly on the connector for removal, as shown.

If unable to disconnect the connector, remove the turbocharger control valve from the mounting bracket for better access to the connector.



Note : The turbocharger control valve bracket may need to be removed first if there is not sufficient room to remove the two mounting capscrews.

Remove the two mounting capscrews.



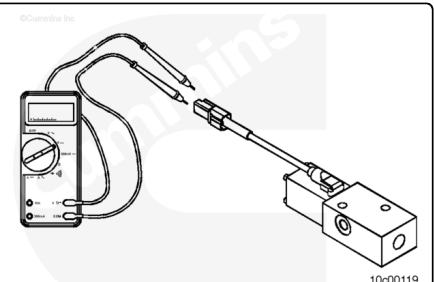
Inspect for Reuse

High Mount

Check the resistance of the turbocharger control valve solenoid.

The resistance **must** be 5 to 9 ohms.

Check the thread holes for damaged threads, contamination, and water entrapment. Clean the thread holes if contamination or water entrapment is evident. Replace the control valve if the



solenoid resistance does **not** meet the specification.

Note : If the turbocharger control valve is replaced, replace the turbocharger air shutoff valve filter.

Refer to Procedure 010-114 (Turbocharger Air Shutoff Valve) in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-114-tr.html>)

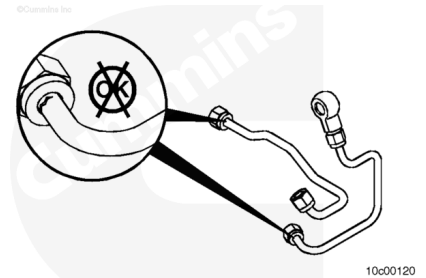
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Check the air lines for cracks. Replace the air lines if cracks are found.

Check the air lines for contamination or water.

Use compressed air to clean the air lines.



Low Mount

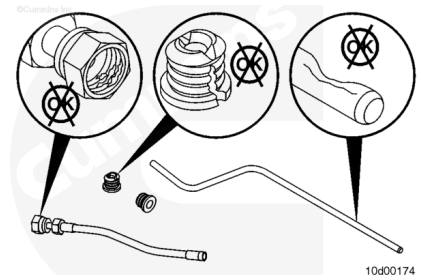
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Check the air lines for cracks. Replace the air lines if cracks are found.

Check the air lines for contamination or water. Use compressed air to clean the air lines.

Check the control valve for water contamination. If water is evident, the control valve should be cycled to purge the water and to verify proper operation. Use the Turbocharger Actuator Test in the INSITE™ electronic service tool to cycle the control valve by extending and retracting the turbocharger actuator at least 10 times.

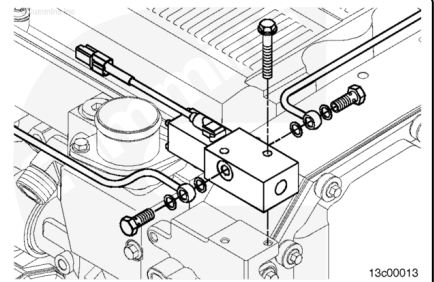


Install

High Mount

⚠ CAUTION ⚠

Do not use any type of thread sealant or thread tape on air lines. The sealant can cause contamination or malfunction of the turbocharger control valve.



Install the two mounting capscrews.

Torque Value: 26 n•m [221 in-lb]

Connect the electrical harness to the turbocharger control valve.

Install the air line from the turbocharger actuator to the turbocharger control valve. Use new sealing washers.

Torque Value: 14 n•m [124 in-lb]

Install the air supply line to the turbocharger control valve. Use new sealing washers.

Torque Value: 35 n•m [26 ft-lb]

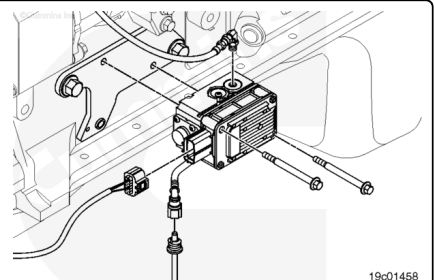
Start and operate the engine. Verify proper operation.

Check for air leaks.

Low Mount

⚠ CAUTION ⚠

Do not use any type of thread sealant or thread tape on air lines. The sealant can cause contamination or malfunction of the turbocharger control valve.



Polymer Body Fittings	8 to 9 n.m	[71 to 80 in-lb]
Aluminum Body Fittings	15 n.m	[133 in-lb]

Install the two turbocharger control valve mounting capscrews.

Torque Value: 15 n•m [133 in-lb]

Note : If the turbocharger control valve bracket was removed, install the bracket and tighten the mounting capscrews.

Torque Value: 80 n•m [59 ft-lb]

Connect the electrical harness to the turbocharger control valve.

Install the air line from the turbocharger actuator to the turbocharger control valve.

Torque Value: 16 n•m [142 in-lb]

Install the OEM air supply line to the turbocharger control valve.

Start and operate the engine. Verify proper operation.

Check for air leaks.